

WARNING: THIS REVIEW IS NOT MEANT TO BE REFLECTIVE OF THE ACTUAL EXAM. THIS IS SIMPLY A SET OF SOME EXTRA PROBLEMS ON THE EXAM MATERIAL. IT IS NOT EXHAUSTIVE OF ALL TYPES OF MATERIAL OR QUESTIONS. YOU SHOULD USE THIS AS ONE STUDY TOOL, NOT YOUR ONLY STUDY TOOL!

MCQs Solution

- 1-b) The status quo or no effect
- 2-c) Independent samples t-test
- 3-b) Contains the true population parameter with a certain level of confidence
- 4-a) Reject the null hypothesis
- 5-c) All of the above
- 6-b) Failing to reject the null hypothesis when it is false
- 7-a) There is a 95% chance the population mean lies within the interval
- 8-c) Set the significance level after data analysis
- 9-a) Sample size and confidence level
- 10-a) Rejecting the null hypothesis when it is false

Hypothesis Testing Problems Solution

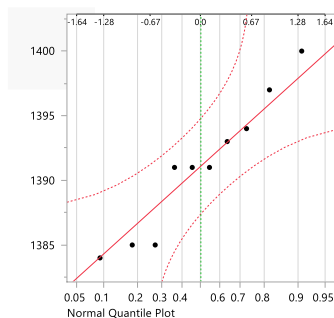
Problem: 11

- 1. $\alpha = 0.05$
- 2. Two Sample T-test
- 3. $H_0: \mu_{day1} - \mu_{day2} = 0$
 $H_a: \mu_{day1} - \mu_{day2} \neq 0$
- 4. $t_0 = -2.461576$
- 5. Pvalue: 0.0252
- 6. Pvalue = 0.0252 < alpha = 0.05, I will reject H_0 .
- 7. Can conclude the average amount is changing between these two days.

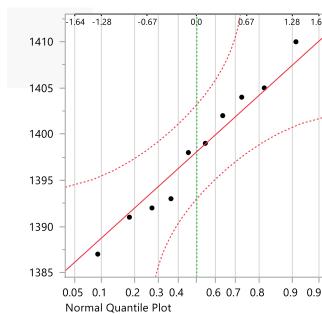
Assume
First and Second data approx. normal

Check
See below. Both plots look ok.

First Day



Second Day

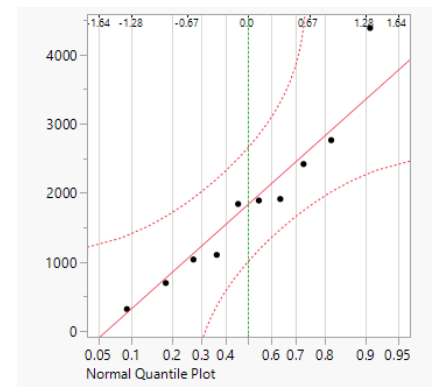


Problem: 12

1. $\alpha = 0.10$
2. Paired Data T-test
3. $H_0: \mu_D = 0$ (difference = 6th - 13th)
 $H_a: \mu_D > 0$
4. $t_0 = 4.93643$
5. Pvalue = 0.0004
6. Pvalue = 0.0004 < alpha = 0.10 → Reject H_0
7. Can conclude there was less traffic on the streets on Friday the 13th.

Assume
RS
Approx normal differences

Check
ask
See plot to right. Looks fine.



Problem: 13

1. $\alpha = 0.05$
2. Test for Single Proportion
3. $H_0: p = 0.05$
 $H_a: p < 0.05$
4. NONE
5. Pvalue = 0.1569 (depends on rounding for test statistic)
6. Pvalue = 0.1569 > alpha = 0.05 → Fail to reject H_0
7. Cannot conclude the percent of nonconforming bricks is less than 5%.

Problem: 14

1. $\alpha = 0.05$
2. Z test for Single Mean
3. $H_0: \mu = 16$
 $H_a: \mu > 16$
4. $z_0 = 6.8872$
5. Pvalue < 0.0001
6. Pvalue $< 0.0001 < \alpha = 0.05 \rightarrow$ Reject H_0
7. Can conclude the mean weight is greater than 16 ounces.

No assumptions: all conditions met in problem

Problem: 15

1. $\alpha = 0.10$
2. T test for Single Mean
3. $H_0: \mu = 3.0$
 $H_a: \mu > 3.0$
4. $t_0 = 1.0195$
5. Pvalue = 0.1626
6. Pvalue = 0.1626 $> \alpha = 0.10 \rightarrow$ Fail to reject H_0
7. Cannot conclude the average GPA is greater than 2.7.

Assume
RS
Approx normal population
Doesn't look normal!

Check
ask/research
See plot to right.

